

recon

User Manual

Version 0.63.0

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Repository · <https://github.com/thomas-starweb/recon>

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About this manual

recon is a versatile network reconnaissance CLI written in Rust. It started as a curl clone and grew into a multi-protocol investigation tool covering HTTP(S), TLS certificate inspection, DNS, WHOIS, ping, traceroute, barcode encoding and decoding, file compression and archiving, markdown/HTML/PDF conversion, and a full Rhai script engine that exposes every protocol probe and helper for automation.

This manual covers every user-facing flag, every script binding, and shows how to combine them. The built-in `recon --help <topic>` command remains the quick reference; this document is the long-form companion.

A PDF rendering of this manual lives alongside it at [MANUAL.pdf](#). It is regenerated every time the markdown changes — see `CLAUDE.md` for the maintenance policy. Generate locally with:

```
recon --md-to-pdf docs/MANUAL.md \  
  --toc --toc-depth 3 --gfm --unsafe-html --page-break-on-h1 \  
  --doc-title 'recon User Manual' \  
  -o docs/MANUAL.pdf
```

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The table of contents above is auto-generated from H1/H2/H3 headings. Every entry is a clickable anchor in the PDF. For a narrative walkthrough of the structure, the four parts are:

- **Part I — Getting started:** introduction, installation, quick start.
 - **Part II — CLI reference:** every flag grouped by area, with examples.
 - **Part III — Script engine:** the Rhai interpreter, every binding, many examples.
 - **Part IV — Appendices:** exit codes, env vars, config, glossary.
-

Introduction

recon is a single-binary command-line tool. Its design goals, in rough order:

1. **Curl-compatible** where it overlaps with curl. Flag names, short forms, and behaviours match curl for HTTP(S) requests so existing tooling keeps working. `recon --help curl` lists the compatibility status.
 2. **Pure Rust** where possible. request + rustls for HTTP(S); hickory for DNS; comrak for markdown; flate2/brotli/zstd for compression; rxing for barcodes; tungstenite for WebSockets; rumqttn for MQTT. Where a pure-Rust path doesn't exist (headless-Chrome PDF rendering, for instance), recon shells out to a named external CLI (`agent-browser`) rather than linking the external dependency.
 3. **Protocol-rich**. HTTP(S), FTP(S), SFTP, TFTP, Gopher, SMTP(S), POP3(S), IMAP(S), SSH, SCP, Telnet, LDAP(S), WS(S), RTSP(S), Dict, NTP, Memcached, Redis, MQTT(S), IPFS/IPNS, Unix sockets, plus ping/traceroute.
 4. **Scriptable**. A Rhai script engine exposes every protocol probe, plus cryptography, compression, barcodes, JSON/YAML/XML handling, SQLite, file I/O, and multi-threaded concurrency. Scripts can stand in for shell pipelines and integration-test harnesses.
 5. **Diagnostic-friendly**. Verbose output, curl-style write-out, prettified JSON/XML, precise error messages, exit codes that match curl's.
-

Installation

From source

```
git clone https://github.com/thomas-starweb/recon
cd recon
cargo build --release
cp target/release/recon ~/.local/bin/      # or /usr/local/bin with sudo
```

You'll need:

- Rust 1.75 or newer (`rustup install stable`).
- A system linker; `cargo` handles the rest.
- For the optional `--html-to-pdf` / `--md-to-pdf` features: `agent-browser` on `$PATH` (`brew install agent-browser` on macOS, or `npm install -g agent-browser` elsewhere).

First-run bootstrap

```
recon --init
```

creates `~/.recon/` with `script/`, `jars/`, `sni/`, and a commented `config.toml` skeleton. Existing files are never overwritten. See the [~/.recon/ layout](#) appendix.

Quick start

```
# HTTP(S)
recon https://api.example.com/status
recon -X POST https://api.example.com/users -d '{"name":"a"}' -H 'Content-Type: application/json'
recon --json '{"q":"rust"}' https://api.example.com/search # auto-sets headers
recon -L https://short.url/foo # follow redirects
recon https://site.example.com --LHEAD # follow + headers

# Inspection
recon https://example.com --cert # inspect TLS
recon example.com --dns # A/AAAA/MX records
recon --ping 8.8.8.8 # TCP + ICMP
recon --traceroute 8.8.8.8
recon --whois example.com

# Email-protection
recon example.com --spf --dmarc --dkim selector1 --mta-sts --bimi --tls-rpt

# Barcodes
recon --encode qr -o out.png 'https://example.com'
recon --decode out.png # → qr<TAB>

# Conversions
recon --md-to-pdf README.md --toc --gfm -o README.pdf
recon --compare a.json b.json # GNU-diff

# Scripting
recon --script my-probe.rhai https://api.example.com
recon --init # bootstrap

# Help surface
recon --help # clap-generated
recon --flags # curl-style
recon --help http # deep-dive
recon --help script # deep-dive
recon --examples # curated examples
recon --version # protocols
```

How recon is organized

recon has one binary with many modes. The mode is selected by flag:

Mode group	Selector flags
HTTP request (default)	no mode flag, or a URL positional argument
Source-inspection	<code>--hash</code> , <code>--compress</code> , <code>--decompress</code> , <code>--encode</code> , <code>--decode</code> , <code>--encrypt</code> , <code>--decrypt</code>
Email-protection	<code>--spf</code> , <code>--dmarc</code> , <code>--dkim</code> , <code>--mta-sts</code> , <code>--bimi</code> , <code>--tls-rpt</code>
Network tests	<code>--ping</code> , <code>--traceroute</code> , <code>--netstatus</code> , <code>--whois</code>
Certificate inspection	<code>--cert</code> , <code>--dns</code> , or positional URL with <code>--cert</code> / <code>--dns</code>
SMTP send	<code>--mail-from</code> and <code>--mail-to</code>
Serve	<code>--serve</code> , <code>--serve-tls</code>
JWT	<code>--jwt-view</code> , <code>--jwt-sign</code> , <code>--jwt-validate</code>
Archive	<code>--archive</code> , <code>--extract</code>
Checkdigit	<code>--checkdigit</code> , <code>--checkdigit-create</code> , <code>--checkdigit-list</code>
Sample	<code>--sample</code> , <code>--sample-list</code>
Browser	<code>--browser-screenshot</code>
Compare	<code>--compare</code>
Docs	<code>--md-to-html</code> , <code>--md-to-pdf</code> , <code>--html-to-pdf</code>
Script	<code>--script</code> (turns recon into a Rhai interpreter)
Meta	<code>--init</code> , <code>--editor-cleanup</code> , <code>--list-charsets</code> , <code>--iconv</code>

Only the modes in the first group do an HTTP request. Every other mode is stand-alone.

Global conventions

- **-h, --help** prints the clap-generated flag summary. **--help <topic>** routes to a deep-dive help topic (see `recon --help` footer for the list).
 - **-V, --version** prints the curl-compatible multi-line banner. **--version-short** prints just the version number.
 - **-v** increases verbosity. Repeat (**-vv** , **-vvv**) for more detail. At default, connection lines and protocol handshakes are suppressed.
 - **-s, --silent** suppresses progress + verbose output. **--show-error** re-enables error output when **-s** is set.
 - **-o <FILE>** writes the response body to a file instead of stdout. **-O** / **--remote-name** derives the filename from the URL's path. **-J** / **--remote-header-name** respects Content-Disposition.
 - **-f, --fail** exits non-zero on HTTP 4xx/5xx (no body printed). **--fail-with-body** prints the body but still exits non-zero.
 - **-k, --insecure** disables TLS cert verification. Use for self-signed tests; never in production scripts.
 - **-p, --pretty** pretty-prints JSON / XML / YAML bodies. Auto-detection by Content-Type; can be disabled with **--no-pretty**.
 - **-L, --location** follows HTTP redirects. **--max-redirs N** caps the chain (default 10). **--LHEAD** follows and prints each hop's headers.
 - **-A "..."** sets the User-Agent. Default is `recon/<version>`.
 - **-e <URL>** sets the Referer header.
 - **-H 'Name: Value'** adds a request header. Repeat for multiple.
 - **--max-time <SECS>** overall operation timeout. **--connect-timeout <SECS>** TCP connect timeout (default 30).
 - **-w '<FORMAT>'** writes a curl-compatible write-out string to stdout after the body. See [Write-out format](#).
 - **--json <DATA>** curl's **--json** compatibility: sends DATA as a JSON body and auto-sets `Content-Type: application/json` + `Accept: application/json`.
 - **-T <FILE>** uploads the file as the request body (default method PUT).
 - **Environment variables** — many flags honor the same env var curl would (`HTTP_PROXY` , `HTTPS_PROXY` , `NO_PROXY` , `ALL_PROXY` , `SSL_CERT_FILE` , `CURL_CA_BUNDLE`). `recon`-specific ones are prefixed `RECON_`. See [Environment variables](#).
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Part II — CLI reference

HTTP / HTTPS requests

The default mode. A positional URL (or `--url <URL>`) triggers an HTTP request through the request + rustls stack.

0.62.0 curl-parity additions — see the curl easy-wins section of `recon --examples` and the relevant flags table rows below. Quick summary: `-r/--range`, `-z/--time-cond`, `--etag-compare`, `--etag-save`, `--timestamping`, `--max-filesize`, `--url-query`, `--disallow-username-in-url`, `--remove-on-error`, `--no-clobber`, `--create-file-mode`, `-N/--no-buffer`, `-D/--dump-header`, `--stderr`, `--no-progress-meter`, `--tcp-nodelay`, `--no-keepalive`, `--keepalive-time`, `--capath`, `--ca-native`, `--tls-max`, `--connect-to`, `--oauth2-bearer`, `--xattr`, `--spider`.

Core flags

Flag	Description
<code>-X, --request <METHOD></code>	GET, POST, PUT, PATCH, DELETE, HEAD. Default GET (PUT when <code>-T</code> set, POST when <code>-d</code> set).
<code>-H, --header <NAME: VALUE></code>	Add a request header. Repeatable.
<code>-d, --data <BODY @FILE></code>	Request body. <code>@file</code> reads from disk, <code>@-</code> reads stdin. Promotes GET → POST unless <code>-G</code> .
<code>--json <DATA></code>	Curl-compatible: sends DATA as JSON, sets <code>Content-Type: application/json + Accept</code> .
<code>--data-raw <DATA></code>	Like <code>-d</code> but <code>@file</code> is NOT processed (sends literal string).
<code>--data-binary <DATA></code>	Like <code>-d</code> but CR/LF are not stripped from <code>@file</code> content.
<code>--data-urlencode <DATA></code>	URL-encode DATA. Repeatable; values joined with <code>&</code> .
<code>-G, --get</code>	Send <code>-d</code> data as query parameters on GET (body empty).
<code>-T, --upload-file <PATH></code>	Upload file as request body. Default PUT.
<code>-L, --location</code>	Follow redirects (3xx).

Flag	Description
<code>--max-redirs <N></code>	Cap redirect chain (default 10).
<code>--LHEAD</code>	Follow redirects and print each hop's headers.
<code>-e, --referer <URL></code>	Set Referer header. <code>--referrer</code> alias accepted.
<code>-A, --user-agent <STR></code>	User-Agent. Default <code>recon/<version></code> .
<code>--compressed</code>	Request + auto-decompress gzip/deflate/brotli/zstd.
<code>--max-time <SECS></code>	Total operation timeout (seconds, fractional OK).
<code>--connect-timeout <SECS></code>	TCP connect timeout (default 30).
<code>-f, --fail</code>	Exit non-zero on HTTP 4xx/5xx, suppress body.
<code>--fail-with-body</code>	Exit non-zero on 4xx/5xx but still print body.
<code>-4, --ipv4 / -6, --ipv6</code>	Force IPv4 / IPv6 resolution.
<code>--resolve <HOST:PORT:ADDR></code>	Override DNS for a specific host:port → addr. Repeatable.

Examples

```

# Simple requests
recon https://httpbin.org/get
recon -X POST https://api.example.com/v1/users -d '{"name":"a"}' -H 'Content-Type: application/json'
recon --json '{"query":"rust"}' https://api.example.com/search

# Body from files / stdin
recon https://upload.example.com -d @payload.json -H 'Content-Type: application/json'
cat body.json | recon -X POST https://api.example.com -d @-

# Upload
recon -T ./report.pdf https://upload.example.com/reports/ # PUT by default
recon -T ./report.pdf https://upload.example.com/ -X POST # override method

# Follow redirects
recon -L https://bitly.example/abc
recon --LHEAD https://example.com/ # show each redirect

# Fail fast
recon -f https://api.example.com/users/42 # exit 22 on first failure
recon --fail-with-body https://api.example.com/users/42 # same but still show body

# Rate and timeout
recon --max-time 10 https://slow.example.com/
recon --connect-timeout 3 --max-time 8 https://flaky.example.com/
recon --limit-rate 500K https://download.example.com/big.bin -o big.bin
recon --speed-limit 10000 --speed-time 30 https://stalling.example.com/big.bin

```

Output control

Flag	Description
<code>-o, --output <FILE></code>	Write body to FILE (otherwise stdout).
<code>-O, --remote-name</code>	Filename = URL's last path segment (percent-decoded).
<code>-J, --remote-header-name</code>	Use Content-Disposition filename when saving.
<code>--create-dirs</code>	Create parent dirs for <code>-o</code> path.
<code>-i, --include</code>	Print response headers before body.
<code>-I, --head</code>	Print headers only; no body. Implies <code>-X HEAD</code> .
<code>-s, --silent</code>	Suppress progress + verbose output.
<code>--show-error</code>	Re-enable error output even when <code>-s</code> is set.
<code>-v, --verbose</code>	Verbose. Repeatable: <code>-v</code> , <code>-vv</code> , <code>-vvv</code> .

Flag	Description
<code>-p, --pretty</code>	Pretty-print JSON / XML / YAML bodies.
<code>--no-pretty</code>	Disable auto-pretty (even when content-type suggests it).
<code>-w, --write-out <FORMAT></code>	Print a curl-compatible summary after the body. See Write-out format .
<code>--editor</code>	Open response body in <code>\$EDITOR</code> after the request.
<code>--json</code> (output-side inferred)	Combined with <code>-p</code> , forces JSON pretty-printing regardless of content-type.

Examples

```
# Save to file
recon https://example.com/favicon.ico -O          # → favicon.ico
recon https://example.com/api/users.json -o users.json
recon https://x.example.com/deep/path/file.txt --create-dirs -o downloads/x/f

# Inspect headers
recon -I https://example.com/                     # HEAD
recon -i https://api.example.com/v1/status        # headers + body
recon -i -I https://example.com/                  # forced HEAD

# Pretty-print
recon https://api.example.com/large.json -p
recon https://api.example.com/feed.xml -p         # XML also supported
curl -s https://api.example.com/blob.yaml | recon -p - # stdin source

# Verbose progression
recon -v https://api.example.com/                  # connection + TLS + headers
recon -vv https://api.example.com/                 # plus intermediate request info
recon -vvv https://api.example.com/               # plus raw handshake + wire dump

# Writeout
recon https://api.example.com/ -w '\n%{http_code} in %{time_total}s (%{size_c
```

Authentication & TLS

Flag	Description
<code>-u, --user <USER:PASS></code>	HTTP Basic credentials.
<code>-k, --insecure</code>	Skip TLS cert verification.

Flag	Description
<code>--tlsv1.2 / --tlsv1.3</code>	Force minimum TLS version.
<code>--cacert <PATH></code>	Trust an extra PEM root (on top of system roots).
<code>--interface <IP NAME></code>	Bind outgoing socket to a specific local IP or interface name (eth0, en0 on Linux/macOS via getifaddrs; Windows accepts IP literals only).
<code>--limit-rate <RATE></code>	Throttle download. Suffixes: K, M, G.
<code>--speed-limit <BYTES></code>	Minimum bytes-per-second (fails if rate drops).
<code>--speed-time <SECS></code>	Window for <code>--speed-limit</code> (default 30).

Examples

```
recon -u alice:s3cr3t https://private.example.com/
recon -k https://self-signed.example.com/
recon --tlsv1.3 https://example.com/           # refuse downgrade to 1.2
recon --cacert /etc/corp-root.pem https://internal.corp/
recon --interface 10.0.0.5 https://example.com/ # use a specific source IP
```

Client certificates (mTLS)

Shipped in 0.54.0. Present a client certificate during the TLS handshake for mutual TLS.

Flag	Description
<code>-E, --client-cert <PATH></code>	PEM-encoded client cert. May include the key inline.
<code>--client-key <PATH></code>	PEM key (when cert is cert-only).
<code>--cert-type <PEM DER></code>	Cert format. DER errors with a conversion recipe.
<code>--key-type <PEM DER ENG></code>	Key format. ENG refused (rustls has no engine concept).
<code>--pass <PASS></code>	Placeholder for encrypted PKCS#8 passphrase. Encrypted keys refused with an <code>openssl pkcs8</code> recipe.

Examples

```
# Combined cert+key PEM
recon -E ~/keys/bundle.pem https://mtls.example.com/

# Split cert and key
recon -E client.crt --client-key client.key https://mtls.example.com/

# Decrypt encrypted PKCS#8 externally first
openssl pkcs8 -in client.key.enc -out client.key
recon -E client.crt --client-key client.key https://mtls.example.com/
```

See also: `recon --help client-cert`.

Proxy routing

0.50.0 shipped the full proxy suite. Schemes auto-detected from the URL: `http://`, `https://`, `socks5://`, `socks5h://`.

Flag	Description
<code>-x, --proxy <URL></code>	Proxy URL. Falls back to <code>\$HTTPS_PROXY</code> / <code>\$HTTP_PROXY</code> / <code>\$ALL_PROXY</code> .
<code>-U, --proxy-user <USER:PASS></code>	Basic auth for the proxy.
<code>--noproxy <LIST></code>	Comma-separated bypass list (matches <code>\$NO_PROXY</code> semantics; <code>*</code> means bypass all).
<code>--proxy-insecure</code>	Skip cert verification on the TLS-to-proxy connection.
<code>--proxy-cacert <PATH></code>	Extra CA for the proxy connection.

Examples

```

recon -x http://proxy.corp:3128 https://api.example.com/
recon -x socks5h://127.0.0.1:9050 https://example.onion/
recon -x https://proxy.example.com:8443 -U alice:s3cr3t https://example.com/
HTTPS_PROXY=http://proxy.corp:3128 NO_PROXY='.internal' recon https://api.exa

# Script equivalent
recon --script - <<'EOF'
http("https://api.example.com/", #{
  proxy: "socks5h://127.0.0.1:9050",
  proxy_user: "alice:s3cr3t",
  noproxy: ".internal",
});
EOF

```

Unix-domain sockets

0.51.0. Route the HTTP request over a Unix socket instead of TCP. Host header and path are preserved; only the transport changes.

Flag	Description
<code>--unix-socket <PATH></code>	Socket path.

Examples

```

# Docker API
recon --unix-socket /var/run/docker.sock http://localhost/_ping
recon --unix-socket /var/run/docker.sock -p http://localhost/v1.40/version
recon --unix-socket /var/run/docker.sock http://localhost/v1.40/containers/js

# systemd-activated service
recon --unix-socket /run/app.sock http://localhost/api/v1/status

```

HSTS persistent cache

0.52.0. A curl-compatible HSTS cache that upgrades `http://` to `https://` before sending (when the host has a non-expired cache entry) and updates itself from `Strict-Transport-Security` response headers.

Flag	Description
<code>--hsts <FILE></code>	Load + update the HSTS cache at this path.

Examples

```
# Populate from an https:// response
recon --hsts ~/.recon/hsts.txt https://www.cloudflare.com/

# Future http:// requests to the same host are auto-upgraded
recon --hsts ~/.recon/hsts.txt http://www.cloudflare.com/
# * HSTS: upgrading http:// to https:// for www.cloudflare.com

# File format matches curl's --hsts:
cat ~/.recon/hsts.txt
# # HSTS cache (recon 0.58.1)
# # host expires_unix (leading . = includeSubDomains)
# .www.cloudflare.com 1808493843
```

IPFS / IPNS

0.49.0. `ipfs://<cid>` and `ipns://<name>` URLs are rewritten to an HTTP gateway URL before the request fires.

Flag	Description
<code>--ipfs-gateway <URL></code>	Gateway base URL. Default <code>https://ipfs.io</code> . Also read from <code>\$RECON_IPFS_GATEWAY</code> .

Examples

```
recon ipfs://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG
recon --ipfs-gateway http://127.0.0.1:8080 ipfs://bafy... # local Kubo node
recon ipns://example.eth # via the default
```

Cookies

Flag	Description
<code>-b, --cookiejar <PATH></code>	SQLite cookie jar at PATH (created on demand).
<code>--cookies</code>	Read-only listing of the current cookie jar.
<code>--cookie-set <NAME=VAL></code>	Set a cookie in the jar.
<code>--cookie-delete <NAME></code>	Remove by name.

Cookies persist across invocations when you pass the same `--cookiejar` path. The `.db` format is SQLite; inspect with `sqlite3` or any sqlite viewer.

Examples

```
# Login, save cookies, browse with them
recon https://example.com/login -d 'user=alice&pass=secret' --cookiejar ~/jar
recon https://example.com/dashboard --cookiejar ~/jar.db -p

# Inspect
recon --cookiejar ~/jar.db --cookies

# Inject manually
recon --cookiejar ~/jar.db --cookie-set 'session=abc123; Domain=.example.com;
```

DNS

Flag	Description
<code>--dns</code>	Resolve A, AAAA, MX, TXT, NS, CNAME, SOA for the URL's host.
<code>--dns-type</code> <code><LIST></code>	Comma-separated subset: <code>A, AAAA, MX, TXT, NS, CNAME, SOA, PTR, CAA, DS, DNSKEY, HTTPS, SRV, SVCB</code> .
<code>--dns-servers</code> <code><LIST></code>	Comma-separated resolver IPs.
<code>--dns-ipv4-addr</code> <code><IP></code>	Bind DNS queries to a specific local IPv4.
<code>--dns-ipv6-addr</code> <code><IP></code>	Same for IPv6.
<code>--dns-interface</code> <code><NAME></code>	Accepted at the CLI; not yet plumbed (see OUT-OF-SCOPE.md). Use <code>--dns-ipv4-addr</code> / <code>--dns-ipv6-addr</code> as a workaround.

Examples

```
recon example.com --dns
recon example.com --dns-type A,AAAA,MX
recon example.com --dns --dns-servers 1.1.1.1,8.8.8.8
recon example.com --dns --dns-servers 127.0.0.1:5353 # local resolver
```

TLS certificate inspection

Standalone mode selected by `--cert` (without a `--cert <PATH>` value — it's the bool form of the flag).

Flag	Description
<code>--cert</code>	Inspect the server cert: subject, issuer, SANs, NotBefore/After, fingerprints.
<code>--cert-chain</code>	Walk the full intermediate chain.
<code>--sni</code> <code><HOSTNAME></code>	Override the SNI value sent in the handshake.

Examples

```
recon https://example.com --cert
recon https://example.com --cert --cert-chain
recon https://example.com --cert --sni alt.example.com      # request cert
```

Ping, traceroute, netstatus

Stand-alone modes.

Flag	Description
<code>--ping <HOST></code>	TCP ping by default, ICMP with <code>--icmp</code> .
<code>--icmp</code>	Force ICMP (needs raw sockets; root on Linux, CAP_NET_RAW elsewhere).
<code>--ping-count <N></code>	Packets (default 4).
<code>--traceroute</code> <code><HOST></code>	Same as <code>--ping HOST --traceroute</code> .
<code>--max-hops <N></code>	Hop limit (default 30).
<code>--netstatus</code>	Run the probe bundle defined in <code>~/.recon/config.toml</code> [netstatus] .

Examples

```
recon --ping 8.8.8.8
recon --ping example.com --ping-count 10
recon --ping example.com:443 --icmp          # ICMP with explicit port hint
recon --traceroute 8.8.8.8 --max-hops 20
recon --netstatus                          # connectivity sweep
```

Email protection

A single URL can be probed for its whole email-auth stack in one invocation.

Flag	Description
<code>--spf</code>	SPF record validation (recursive include/redirect, lookup limit)
<code>--dmarc</code>	DMARC policy + record
<code>--dkim <SELECTOR></code>	DKIM for the given selector. Repeatable.
<code>--mta-sts</code>	MTA-STS policy
<code>--bimi [SELECTOR]</code>	BIMI record + optional VMC / SVG fetch
<code>--tls-rpt</code>	SMTP TLS-RPT record

Examples

```
# Full sweep
recon example.com --spf --dmarc --dkim selector1 --mta-sts --bimi --tls-rpt

# Single checks
recon example.com --spf
recon example.com --dkim selector1 --dkim selector2
recon example.com --bimi default          # explicit selector
```

SMTP

Stand-alone SMTP client. Send mail or just probe capabilities.

Flag	Description
<code>--mail-from <ADDR></code>	Envelope sender. Required for send.
<code>--mail-to <ADDR></code>	Envelope recipient. Repeatable.
<code>--mail-subject <STR></code>	Subject header. Default "recon SMTP test".
<code>--mail-body <STR></code>	Body. Accepts @file , @- .

Flag	Description
<code>--mail-header <H: V></code>	Extra header. Repeatable.
<code>--smtp-auth <USER:PASS></code>	AUTH PLAIN → LOGIN chain.
<code>--smtp-helo <NAME></code>	HELO/EHLO name. Default <code>recon.local</code> .
<code>--no-starttls</code>	Don't negotiate STARTTLS.
<code>--dkim-key <PATH></code>	DKIM-sign outbound with this PEM key.
<code>--dkim-selector <SEL></code>	DKIM selector.
<code>--dkim-domain <DOM></code>	DKIM domain (defaults to <code>--mail-from</code> domain).

Examples

```
# Probe only
recon smtp://mail.example.com:25

# Send
recon smtp://smtp.example.com:587 --mail-from me@example.com --mail-to you@example.com
--mail-subject 'Hi' --mail-body 'Test' --smtp-auth me:s3cr3t

# With DKIM signing
recon smtp://smtp.example.com:587 --mail-from me@example.com --mail-to you@example.com
--dkim-key ~/.dkim/default.pem --dkim-selector default
```

Mail retrieval (POP3, IMAP)

Flag	Description
<code>--stls</code>	POP3: upgrade via STLS after CAPA.
<code>--imap-peek</code>	IMAP: use BODY.PEEK (don't flip \Seen).

URLs: `pop3://`, `pop3s://`, `imap://`, `imaps://`.

Examples

```
recon pop3s://alice:s3cr3t@pop.example.com/
recon pop3://alice:s3cr3t@pop.example.com/ --stls
recon imaps://alice:s3cr3t@imap.example.com/INBOX
recon imaps://alice:s3cr3t@imap.example.com/INBOX/3 # fetch message 3
recon imaps://alice:s3cr3t@imap.example.com/INBOX --imap-peek
```


File transfer

0.47.0 shipped FTP/FTPS, SFTP, TFTP, Gopher.

Protocol	URL scheme	Notes
FTP / FTPS	ftp:// , ftps://	Anonymous by default; userinfo in URL or <code>-u</code> .
SFTP	sftp://	SSH-backed. <code>--privkey</code> for a specific key.
TFTP	tftp://	RFC 1350. <code>--tftp-blksize</code> for RFC 2348 block-size negotiation.
Gopher	gopher:// , gophers://	Selector fetch.

Examples

```
# FTP – listing vs fetch
recon ftp://ftp.example.com/                                # directory listing
recon ftp://ftp.example.com/incoming/file.bin -o local.bin   # retrieve file
recon ftp://alice:s3cr3t@ftp.example.com/                   # authentication

# SFTP
recon sftp://me@example.com/home/me/data.csv -o data.csv
recon sftp://me@example.com:2222/srv/ --privkey ~/.ssh/ci_rsa

# TFTP
recon tftp://tftp.example.com/pxelinux.cfg/default
recon tftp://tftp.example.com/boot.img -o boot.img --tftp-blksize 8192

# Gopher
recon gopher://gopher.floodgap.com/
recon gopher://gopher.floodgap.com/0/gopher/proxy
```

Other protocols

SSH, SCP

```
recon ssh://me@example.com -- uname -a                      # SSH exec; prints stdout
recon scp://me@example.com/etc/motd -o motd                 # SCP fetch
```

Telnet

```
recon telnet://router.example.com:23
```

LDAP

```
recon ldap://ldap.example.com -H 'Bind: cn=admin,dc=example,dc=com'
recon ldaps://ldap.example.com # implicit-TLS
```

WebSocket

```
recon wss://echo.websocket.events # handshake + Ping/Pong
```

RTSP

```
recon rtsp://camera.example.com/stream1 # OPTIONS
```

Dict (RFC 2229)

```
recon dict://dict.org/d:recon
```

NTP, Memcached, Redis, MQTT

```
recon ntp://pool.ntp.org # clock offset + rtt
recon memcache://cache.example.com/?stats # text-protocol stats
recon redis://cache.example.com/ # PING
recon redis://cache.example.com/ -X INFO # arbitrary RESP command
recon mqtt://broker.example.com --mqtt-topic status --mqtt-message 'hello'
recon mqtt://broker.example.com --mqtt-user alice --mqtt-pass s3cr3t ...
```

Source comparison

0.53.0 shipped `--compare`. Diff two sources (URL / path / `-` stdin). HTTP(S) sources flow through the full request pipeline so every request flag (`-H`, `-u`, `-L`, `-k`, cookies, proxy, HSTS) applies.

Flag	Description
<code>--compare <A> </code>	Two sources.
<code>--compare-format <FMT></code>	<code>unified</code> (default), <code>summary</code> , <code>sxs</code> .
<code>--compare-context <N></code>	Unified context lines (default 3).

Exit codes follow GNU `diff`: 0 identical, 1 differ, 2+ source-load error.

Examples

```
recon --compare one.json two.json
recon --compare https://api.example.com/v1/status ./baseline.json
curl -s https://a/ | recon --compare - ./baseline.json
recon --compare a.txt b.txt --compare-format summary
recon --compare a.txt b.txt --compare-format sxs
recon --compare a.json b.json --compare-context 5
```

Document conversions

0.58.0 shipped markdown / HTML / PDF. See also [--help docs](#).

Flag	Description
<code>--md-to-html <SRC></code>	Markdown → HTML via comrak. Output <code>-o PATH</code> or stdout.
<code>--md-to-pdf <SRC></code>	Markdown → PDF (via agent-browser). <code>-o PATH</code> required.
<code>--html-to-pdf <SRC></code>	HTML → PDF (via agent-browser).
<code>--toc</code>	Inject a linkable TOC.
<code>--toc-depth <N></code>	Include headings up to H <code>N</code> (default 3).
<code>--toc-title <STR></code>	TOC heading (default "Contents").
<code>--doc-title <STR></code>	<code><title></code> + PDF metadata title.
<code>--doc-css <PATH></code>	Inline a custom stylesheet.
<code>--no-default-css</code>	Skip bundled default CSS.
<code>--gfm</code>	Enable GitHub-flavored extensions (tables, task lists, strikethrough, autolinks, footnotes, tagfilter).
<code>--unsafe-html</code>	Allow raw HTML passthrough (comrak's <code>unsafe_</code> flag). Needed for cover pages and explicit <code><div class="page-break"></code> markers. Off by default; assume the markdown is trusted when on.

Flag	Description
<code>--page-break-on-h1</code>	Start a new PDF page before every top-level <code>#</code> heading except the first. No visible effect in HTML output (Chrome's printToPDF honours the <code>break-before: page</code> rule).

Examples

```
recon --md-to-html README.md --toc --gfm -o README.html
recon --md-to-pdf CHANGELOG.md --toc --gfm --doc-title 'recon release notes'
recon --html-to-pdf report.html -o report.pdf
curl -s https://example.com/doc.md | recon --md-to-html - --toc -o doc.html
recon --md-to-pdf notes.md --toc --doc-css print.css -o notes.pdf
recon --md-to-pdf notes.md --no-default-css --doc-css corp.css -o notes.pdf

# Book-style: cover page + chapter breaks
recon --md-to-pdf book.md --toc --gfm --unsafe-html --page-break-on-h1 \
    --doc-title 'My Book' -o book.pdf
```

Cover pages

With `--unsafe-html`, raw HTML passes through the markdown parser verbatim. The bundled CSS styles `<div class="cover">` as a full-page centered block with an automatic page break after:

```
<div class="cover">

# My Document

<div class="subtitle">An illustrated guide</div>

<hr>

<div class="version">Version 1.0</div>
<div class="date">2026-04-24</div>
<div class="author">Alice Example</div>

</div>

# First chapter

Body text...
```

The cover block uses a `min-height: 90vh` flex container with centered content. `.subtitle`, `.version`, `.date`, `.author`, and `.meta` child classes pick up smaller, muted styling. A horizontal rule becomes a narrow divider.

Page breaks

Two routes:

1. **--page-break-on-h1** — every top-level `#` heading starts a new PDF page (except the first, which opens the document). No markdown changes required.
2. **Explicit markers (needs `--unsafe-html`)** — embed one of:

```
<div class="page-break"></div>
<hr class="page-break">
```

anywhere in the markdown. The CSS `break-after: page` kicks in.

Chrome's printToPDF is the renderer for both `--md-to-pdf` and `--html-to-pdf`, so any CSS that speaks `break-before`, `break-after`, or `page-break-*` works. `@page` rules for size and margins (defined in the bundled default CSS) are honored too — override via `--doc-css` or inline `<style>` with `--unsafe-html`.

Barcode encoding & decoding

Flag	Description
<code>--encode</code> <code><FORMAT></code>	qr, datamatrix, code128, code39, ean13, upca, aztec, pdf417.
<code>--encode-format</code> <code><FMT></code>	ascii (default), svg, png. Inferred from <code>-o</code> extension.
<code>--from-file</code> <code><PATH></code>	Encode input from file (mutually exclusive with positional).
<code>--encode-list</code>	List all supported formats.
<code>--qr-level</code> <code><L M Q H></code>	QR error-correction level (default M).
<code>--hrt / --no-hrt</code>	Human-readable text under 1D barcodes. Default on for EAN/UPC, off for Code128/39. SVG + ASCII only (PNG HRT deferred).
<code>--decode</code> <code><IMAGE></code>	Scan a PNG/JPEG/WebP for any supported format. <code>-</code> for stdin.
<code>--decode-hints</code> <code><LIST></code>	Comma-separated format restriction.
<code>--decode-all</code> <code><IMAGE></code>	Scan for every barcode in the image; one line per detection.

Examples

```
# Encode
recon --encode qr -o qr.png 'https://example.com'
recon --encode qr --qr-level H -o durable.png 'sticker text'
recon --encode pdf417 -o id.png 'stacked linear code'
recon --encode aztec -o ticket.png 'transit ticket'
recon --encode ean13 --encode-format svg '4006381333931' > product.svg
recon --encode qr --from-file msg.txt -o qr-of-file.png

# Decode
recon --decode ticket.png # → aztec<TAB>transit tic
cat code.png | recon --decode -
recon --decode mystery.png --decode-hints qr,datamatrix
recon --decode bottle.jpg --decode-hints ean13
recon --decode-all sheet.png # every code in the image
```

HRT (human-readable text under 1D barcodes)

EAN-13 and UPC-A get HRT by default; Code128 and Code39 don't unless you pass `--hrt`. Implemented for ASCII and SVG output. PNG output ignores the flag — PNG HRT is deferred pending bundled-font work.

```
recon --encode ean13 --encode-format svg '4006381333931' -o retail.svg
recon --encode code128 --hrt --encode-format svg 'SHIP-4711' -o box.svg
recon --encode ean13 --no-hrt '4006381333931' -o bare.svg
```

Hashing

Flag	Description
<code>--hash</code> <code><ALGO></code>	md5, sha1, sha224, sha256, sha384, sha512, sha3_256, sha3_512, blake3.
<code>--hash-list</code>	List supported algorithms.

Examples

```
recon --hash sha256 /etc/hosts
recon --hash sha256 https://example.com/file.iso # hash a URL
cat payload.bin | recon --hash blake3 -
echo -n "hello" | recon --hash md5 -
```

Compression & archiving

Flag	Description
<code>--compress <ALGO></code>	gzip, deflate, brotli, zstd, bzip2, lz4, xz, snap.
<code>--decompress <ALGO></code>	Same set; auto-detect via magic bytes if ALGO is <code>auto</code> .
<code>--compress-list</code>	List supported algorithms.
<code>--compression-level <N></code>	1..22 (depends on algo).
<code>--archive <DEST></code>	Create zip/tar/tar.gz/tar.xz/tar.bz2. Positional args after DEST are sources.
<code>--extract <SRC></code>	Extract an archive. <code>-o DIR</code> to choose destination.

Examples

```
recon --compress gzip /etc/hosts -o hosts.gz
recon --decompress auto hosts.gz -o hosts.plain
recon --compress zstd --compression-level 19 big.bin -o big.zst

recon --archive backup.tar.gz ~/Documents ~/src
recon --extract backup.tar.gz -o restore/
recon --extract release.zip
```

Encryption

age-based + PGP shell-out.

Flag	Description
<code>--encrypt</code>	Encrypt stdin / file to age format. Requires <code>--recipient</code> or <code>--pgp-recipient</code> .
<code>--decrypt</code>	Decrypt age / PGP. Needs <code>--identity</code> or a configured GPG keyring.
<code>--encrypt-keygen</code>	Generate a new age keypair.
<code>--recipient <KEY></code>	Age recipient (X25519 or ssh-ed25519). Repeatable.
<code>--identity <PATH></code>	Age identity file for decryption.
<code>--passphrase</code>	Use passphrase-based (scrypt) encryption. Mutually exclusive with <code>--recipient</code> .
<code>--armor</code>	Armored (base64 PEM) output.

Flag	Description
<code>--pgp-recipient <EMAIL FPR></code>	PGP recipient (shells out to <code>gpg</code>).

Examples

```
# Generate
recon --encrypt-keygen > age.key          # public key printed to stderr

# Encrypt
recon --encrypt --recipient age1abc... README.md -o README.md.age
recon --encrypt --passphrase secret.txt -o secret.txt.age --armor

# Decrypt
recon --decrypt --identity age.key README.md.age -o README.md

# PGP
recon --encrypt --pgp-recipient alice@example.com msg.txt -o msg.txt.pgp
```

Check digits

Flag	Description
<code>--checkdigit <ALGO></code>	Verify an input.
<code>--checkdigit-create <ALGO></code>	Compute and append the check digit.
<code>--checkdigit-list</code>	List supported algorithms (70+).

Notable algorithms shipped in 0.61.0:

- **Latin-American + Australian + Mexican tax IDs:** `br_cpf`, `br_cnpj`, `ar_cuit`, `ar_cuil`, `cl_rut`, `pe_ruc`, `au_abn`, `mx_rfc`. All except ABN support `--checkdigit-create`; ABN's mod-89 algorithm has no single-digit inverse.
- **110+ year warnings** on Nordic + Bulgarian personal IDs: `cpr` (Denmark), `henkilotunnus` (Finland), `fodselsnummer` (Norway), `bg-egn` (Bulgaria). Same idiom as the Swedish `personnummer`: when the parsed birth year implies age ≥ 110 , the verdict's `comment` flags a likely data-entry error.

Examples


```
recon --checkdigit isbn13 '9780131103627'      # → valid
recon --checkdigit-create isbn13 '978013110362'  # → 9780131103627
recon --checkdigit-create ean13 '400638133393'  # → 4006381333931
recon --checkdigit personnummer '19900101-1239' # Swedish personal ID
recon --checkdigit-list | head -20
```

Sample data

Flag	Description
<code>--sample <NAME></code>	Emit a sample payload (faker-style).
<code>--sample-list</code>	List all named samples.

Examples

```
recon --sample-list
recon --sample json-user
recon --sample json-user -o user.json
recon --sample css                      # minimal CSS boilerplate
```

JWT tokens

Flag	Description
<code>--jwt-view <TOKEN></code>	Decode + pretty-print. No signature check.
<code>--jwt-sign <PAYLOAD></code>	Sign a JWT. Pair with <code>--jwt-secret</code> or <code>--jwt-key</code> .
<code>--jwt-validate <TOKEN></code>	Validate signature + standard claims.
<code>--jwt-secret <STR></code>	HMAC secret (HS256/384/512).
<code>--jwt-key <PATH></code>	RSA/ECDSA/Ed25519 PEM key.
<code>--jwt-alg <ALG></code>	HS256, HS384, HS512, RS256, RS384, RS512, ES256, ES384, EdDSA.

Examples

```
recon --jwt-view "$(cat token.txt)"
recon --jwt-sign '{"sub":"alice","exp":9999999999}' --jwt-secret sharedkey --
recon --jwt-validate "$TOKEN" --jwt-secret sharedkey
recon --jwt-sign '{"sub":"alice"}' --jwt-key private.pem --jwt-alg RS256
```

Text encoding

Flag	Description
<code>--iconv <SRC:DST></code>	Standalone <code>iconv</code> replacement.
<code>--list-charsets</code>	List supported charset labels.

Examples

```
recon --iconv utf-8:iso-8859-1 greek.txt -o greek-latin1.txt
echo 'Hello' | recon --iconv utf-8:utf-16le - -o hello.utf16
recon --list-charsets | head -20
```

Serve mode

Flag	Description
<code>--serve [ADDR]</code>	Start an HTTP server serving the cwd (default <code>127.0.0.1:8000</code>).
<code>--serve-tls [ADDR]</code>	HTTPS. Requires <code>--serve-cert</code> + <code>--serve-key</code> .
<code>--serve-cert <PATH></code>	Server cert (PEM).
<code>--serve-key <PATH></code>	Server key (PEM).
<code>--serve-sni</code> <code><HOST:CERT:KEY></code>	Per-hostname SNI mapping. Repeatable.

Examples

```
recon --serve
recon --serve 0.0.0.0:3000
recon --serve-tls 0.0.0.0:8443 --serve-cert cert.pem --serve-key key.pem
recon --serve-tls :443 --serve-sni a.example.com:a.pem:a.key --serve-sni b.ex
```

Write-out format

The `-w '<FORMAT>'` flag prints a curl-compatible summary after the response. Variable names match curl's.

Variable	Description
<code>%{http_code}</code>	HTTP status code
<code>%{size_download}</code>	Body size in bytes
<code>%{size_header}</code>	Response header bytes
<code>%{size_request}</code>	Request bytes sent
<code>%{size_upload}</code>	Uploaded bytes
<code>%{speed_download}</code>	Bytes-per-second
<code>%{time_total}</code>	Total operation time (seconds, fractional)
<code>%{time_starttransfer}</code>	TTFB
<code>%{time_redirect}</code>	Time spent on redirects
<code>%{url}</code>	Final URL (after redirects)
<code>%{url_effective}</code>	Same as <code>%{url}</code>
<code>%{content_type}</code>	Response Content-Type
<code>%{num_redirects}</code>	Redirect count
<code>%{remote_ip}</code>	Final peer IP
<code>%{remote_port}</code>	Peer port
<code>%{local_ip} / %{local_port}</code>	Local side
<code>%{scheme}</code>	http or https

Not yet accurate: `time_namelookup`, `time_connect`, `time_appconnect`, `time_pretransfer` render as `0.000000` — see OUT-OF-SCOPE.md.

Examples

```
recon https://example.com/ -w '\n%{http_code} %{time_total}s %{size_download}
recon -I https://example.com/ -w '%{scheme}://%{remote_ip}:%{remote_port} %{r
recon -L https://short.url/x -w '%{num_redirects} hops → %{url_effective}\n'
```

Browser automation

Flag	Description
<code>--browser-screenshot <URL></code>	Open in agent-browser, save a PNG to <code>-o PATH</code> .

Requires `agent-browser` on `$PATH`. See [Script engine → agent-browser](#) for deeper automation.

Examples

```
recon --browser-screenshot https://example.com -o example.png
```

Meta flags

Flag	Description
<code>--flags</code>	Alphabetical curl-style flag listing (<code>--help all</code> equivalent).
<code>--examples</code>	Curated examples, paged.
<code>--init</code>	Bootstrap <code>~/.recon/</code> (script/, jars/, sni/, config.toml).
<code>--editor</code>	Open response in <code>\$EDITOR</code> .
<code>--editor-cleanup</code>	Remove leftover <code>~/.recon/editor-*</code> tempfiles.
<code>--no-pager</code>	Disable paging of <code>--help</code> / <code>--examples</code> / <code>--flags</code> .

`--flags` — the quick lookup

`recon --flags` prints every flag alphabetically sorted by long name, curl's `--help all` layout:

```
(short or 4-space pad) --long <VALUE>  short description
```

Short description is capped at ~52 characters (first sentence of each flag's internal help text). Use this as the quick index when you know roughly what you want but not the flag name; follow up with `recon --help <topic>` for the long-form deep-dive.

Example:

```
$ recon --flags | head
  --age                Force the age backend, even if...
  --archive <DEST>     Create an archive
  --armor              Produce ASCII-armored output...
  --bimi <SELECTOR>     Validate the BIMi record
  --browser-screenshot <URL> Open URL in a browser and save...
  --cacert <PATH>       Path to a PEM-encoded CA certificate...
  --cert               Fetch and display the server's TLS cert
  --cert-type <PEM|DER> Format of --client-cert
-E, --client-cert <PATH> Client certificate for mTLS
  --client-key <PATH>   Private key for the client certificate
```

The listing is auto-paged through `$PAGER`. Pipe to `grep` for search:

```
recon --flags | grep -i cookie
recon --flags | grep -E '^s*-[a-zA-Z],' # only flags with short keys
```

Part III — Script engine

recon ships an embedded [Rhai](#) interpreter. Every protocol probe, every cryptography primitive, every helper is exposed to scripts. Scripts run via `--script PATH`; `return N` from the top level becomes the process exit code.

Running scripts

```
recon --script ./probe.rhai           # run an absolute / relative
recon --script probe                  # falls back to ~/.recon/scr
recon --script probe https://example.com alice  # positional args → args[1],
```

Script lookup order:

1. Exact path (with or without `.rhai` extension).
2. `~/.recon/script/<name>.rhai`.
3. If neither exists — error.

The shipped example scripts in `script/*.rhai` are copy-friendly starting points:

```
cp script/*.rhai ~/.recon/script/
recon --script http example.com
```

Bare-word invocation

After `recon --init` + copying scripts to `~/.recon/script/`, scripts become first-class by name:

```
recon --script tcp-echo 127.0.0.1:9000
recon --script doc-convert README.md output.pdf
recon --script client-cert ~/keys/bundle.pem https://mtls.example.com/
```

Script language (Rhai)

Rhai is a lightweight embedded script language. Key facts for newcomers:

- **Let bindings:** `let x = 42;` (mutable by default; use `const` for compile-time constants).

- **Types:** integers (i64), floats (f64), bools, strings, arrays, maps (object literals: `#{ key: value }`), Blob (Vec).
- **String interpolation:** ``Hello ${name}``.
- **Closures / function pointers:** `|a, b| { a + b }`.
- **For loops:** `for i in 0..5 { ... }` (integer ranges), `for item in array { ... }`, `for (k, v) in map { ... }`.
- **Functions:** `fn f(a) { a * 2 }`.
- **Imports:** `import "module-name" as m;` — resolved against the script's directory and `~/.recon/script/`.
- **Error handling:** `try { ... } catch(e) { ... }`.
- **Reserved words** that might surprise: `spawn` (use `thread_spawn`), `async`, `await`, `throw` (all reserved even when unused).

Useful idioms:

```
// Map with computed keys
let cfg = #{
  user: env("USER"),
  host: "api.example.com",
  timeout_ms: 5000,
};

// Chainable array methods
let evens = (0..20).filter(|n| n % 2 == 0).map(|n| n * 10);

// Early return
if !file_exists("/etc/passwd") { return 1; }
```

CLI inheritance

When a script runs, two read-only constants land at the top of its scope:

- **args** — an array of strings. `args[0]` is the script path; `args[1..]` are the positional arguments after `--script PATH`.
- **flags** — a map mirroring the relevant CLI flags (lower-case, snake_case keys). Useful for scripts that want to respect `-v`, `-k`, `--connect-timeout`, etc. at invocation time.

```
// Reasonable defaults, overridable via args
let url = if args.len() > 1 { args[1] } else { "https://example.com/" };
let timeout = flags.connect_timeout; // from -C / --connect-timeout

let r = http(url, #{ timeout_ms: timeout * 1000 });
print(`${r.status} ${r.url}`);
```

Bindings that take their own opts map (e.g. `http(url, opts)`) layer the caller's opts ON TOP of the CLI defaults. If the caller wants to ignore the CLI's `-H` header list, they can pass `headers: []` in their opts.

Core helpers

Exposed by `src/script/bindings/helpers.rs`. Always available.

Function	Returns	Description
<code>sleep_ms(ms)</code>	—	Block the current thread.
<code>sleep(ms)</code>	—	Alias of <code>sleep_ms</code> (added in 0.56.0 for thread-side readability).
<code>env(name)</code>	string	Process environment variable; <code>""</code> if unset.
<code>env(name, default)</code>	string	With fallback.
<code>now()</code>	int	Unix seconds.
<code>now_ms()</code>	int	Unix milliseconds.
<code>assert(cond, msg)</code>	—	Throws on false.
<code>json_parse(s)</code>	Dynamic	Parse JSON text → native Rhai value.
<code>json_stringify(v)</code>	string	Serialize to compact JSON.
<code>json_stringify(v, pretty)</code>	string	<code>pretty: true</code> = 2-space indent.
<code>json_stringify(v, n)</code>	string	<code>n</code> = spaces per indent (clamped 1..=8).

Rhai's built-in `print(x)` writes `x` + newline via the engine's debug callback. `recon` adds byte-precise siblings (see next section).

Examples


```

// Probe a URL, fail fast
let r = http("https://api.example.com/health");
assert(r.status == 200, `expected 200, got ${r.status}`);

// JSON round-trip
let obj = json_parse(r.body);
print(json_stringify(obj, true));    // pretty

// Time gate
let t0 = now_ms();
let r = http("https://slow.example.com/");
let dt = now_ms() - t0;
print(`took ${dt} ms`);

// Polling with timeout
let deadline = now() + 30;
while now() < deadline {
    let r = http("https://example.com/ready");
    if r.status == 200 { return 0; }
    sleep_ms(500);
}
return 1;

```

Output bindings

0.53.0. Byte-precise output that Rhai's built-in `print()` can't do.

Function	Description
<code>print_raw(s)</code> / <code>print_raw(blob)</code>	Write to stdout without a trailing newline; flush.
<code>eprint(s)</code>	Write to stderr with a newline.
<code>eprint_raw(s)</code> / <code>eprint_raw(blob)</code>	Write to stderr without newline; flush.
<code>flush()</code>	Explicit stdout flush.

Examples

```
// Progress indicator
print_raw("loading");
for i in 0..10 {
    sleep_ms(200);
    print_raw(".");
}
print_raw("\n");

// Send bytes to stdout (e.g. for piping into another tool)
let bytes = http("https://example.com/image.png").body;
print_raw(bytes);

// Log to stderr; keep stdout for data
eprint(`[${now_ms()}] starting probe...`);
print_raw(json_stringify(result));    // stdout stays parseable
```

File I/O bindings

0.53.0. Both whole-file convenience and a streaming handle API.

Whole-file helpers

Function	Description
<code>file_read(path)</code>	Read entire file → Blob.
<code>file_write_all(path, blob str)</code>	Overwrite. Returns bytes written.
<code>file_append_all(path, blob str)</code>	Append; creates if absent.
<code>file_exists(path)</code>	Bool.
<code>file_size(path)</code>	Bytes.
<code>file_delete(path)</code>	Unlink.

`path` accepts a filesystem path or a `file://` URL.

Streaming handle API

Function	Description
<code>file_open(path, mode)</code>	Returns a <code>FileHandle</code> . Modes: <code>r</code> , <code>w</code> (truncate+create), <code>rw</code> , <code>rw+</code> / <code>w+</code> (read+write+create+truncate), <code>a</code> (append+create), <code>ra</code> (append+read).
<code>file_read(h, n)</code>	Read up to <code>n</code> bytes → Blob.
<code>file_read_all(h)</code>	Drain to Blob.

Function	Description
<code>file_write(h, blob str)</code>	Write all bytes. Returns count.
<code>file_seek(h, pos, whence)</code>	whence = <code>start</code> <code>cur</code> <code>end</code> . Returns new position.
<code>file_tell(h)</code>	Current position.
<code>file_flush(h)</code>	Sync buffered writes.
<code>file_close(h)</code>	Close (drops the underlying file).

Examples

```
// Whole-file
let data = file_read("config.json");
file_write_all("/tmp/copy.json", data);
file_append_all("/tmp/audit.log", `probe at ${now()}\n`);

// Check before read
if !file_exists("secrets.key") {
    eprint("missing key");
    return 2;
}

// Streaming – copy with 4KB chunks
let src = file_open("big.bin", "r");
let dst = file_open("/tmp/big.bin.copy", "w");
loop {
    let chunk = file_read(src, 4096);
    if chunk.len() == 0 { break; }
    file_write(dst, chunk);
}
file_close(src);
file_close(dst);

// Random access
let h = file_open("/tmp/log.bin", "rwc");
file_write(h, "record-1\n");
file_write(h, "record-2\n");
file_seek(h, 0, "start");
let head = file_read_all(h);
print(`contents: ${head.len()} bytes`);
file_close(h);
```

Comparison bindings

0.53.0. Diff two in-memory values (strings or Blobs).

Function	Returns	Description
<code>compare(a, b)</code>	Map	<code>{ identical, binary, a_bytes, b_bytes, added, removed, diff }</code> .

Both `a` and `b` accept strings OR Blobs. The binding does NOT fetch URLs — scripts should pre-load via `http()` or `file_read()` and pass the bytes.

Examples

```
// Compare two API responses
let a = http("https://api.v1.example.com/users/42").body;
let b = http("https://api.v2.example.com/users/42").body;
let r = compare(a, b);
if r.identical {
    print("parity OK");
    return 0;
}
print(`diverged: +${r.added} / -${r.removed}`);
print_raw(r.diff);
return 1;

// Compare a live URL against a baseline file
let live = http("https://example.com/status").body;
let baseline = file_read("status.baseline").to_string();
let r = compare(live.to_string(), baseline);
if !r.identical { print_raw(r.diff); }

// Binary content
let old = file_read("logo.v1.png");
let new = file_read("logo.v2.png");
let r = compare(old, new);
print(`binary: ${r.binary}, size delta: ${r.b_bytes - r.a_bytes}`);
```

HTTP binding

The workhorse. Two call forms:

```
http(url)           → response map
http(url, opts)     → response map
```

Response map

Key	Type	Description
status	int	HTTP status code.
url	string	Original URL.
final_url	string	After redirects.
headers	Map	Response headers; values are strings (first) or arrays when repeated.
body	Blob string	Response body. String when content-type is text-ish; Blob otherwise.
duration_ms	int	Total wall-clock time.
http_version	string	HTTP/1.1 , HTTP/2 , etc.

Options map

Every key is optional.

Key	Type	Description
method	string	GET, POST, PUT, PATCH, DELETE, HEAD.
headers	Map	Key → value (or array of values). Merged with CLI <code>-H</code> defaults.
body	string / Blob	Request body.
json	Dynamic	Serialize as JSON, set <code>Content-Type</code> + <code>Accept</code> .
form	Map	URL-encoded form body.
multipart	Array of Maps	Each Map has <code>name</code> , <code>value</code> or <code>file</code> , optional <code>filename</code> , <code>content_type</code> .
basic_auth	string	<code>"user:pass"</code> .
bearer	string	Bearer token.
user_agent	string	Override User-Agent.
referer	string	Referer.
timeout_ms	int	Total operation timeout.
connect_timeout	int	TCP connect timeout (seconds).
insecure	bool	Skip TLS verification.
follow_redirects	bool	Enable / disable redirect follow.

Key	Type	Description
max_redirects	int	Redirect cap.
compressed	bool	Request + auto-decompress.
cookiejar	string	Path to a SQLite cookie jar.
proxy	string	Proxy URL.
proxy_user	string	Proxy basic auth.
noproxy	string	Bypass list.
proxy_insecure	bool	Skip cert verification on proxy.
proxy_cacert	string	Extra CA for proxy.
unix_socket	string	Route via Unix socket.
hsts	string	Path to HSTS cache.
client_cert	string	PEM client cert.
client_key	string	PEM client key.
cert_type / key_type	string	PEM DER ENG (see CLI docs).
pass	string	PKCS#8 passphrase (reserved).

Examples

```

// Simple GET
let r = http("https://api.example.com/status");
print(`${r.status} ${r.body.len()} bytes`);

// POST JSON
let r = http("https://api.example.com/users", #{
    method: "POST",
    json: #{ name: "alice", email: "a@example.com" },
});
assert(r.status == 201, "create failed");

// Form POST
http("https://api.example.com/login", #{
    method: "POST",
    form: #{ user: "alice", pass: "s3cr3t" },
    cookiejar: "/tmp/jar.db",
});

// Multipart upload
http("https://upload.example.com/avatars", #{
    method: "POST",
    multipart: [
        #{ name: "title", value: "profile" },
        #{ name: "image", file: "avatar.png", content_type: "image/png" },
    ],
    bearer: env("API_TOKEN"),
});

// Bearer auth + timeout + compressed
let r = http("https://api.example.com/items", #{
    bearer: env("API_TOKEN"),
    timeout_ms: 5000,
    compressed: true,
    headers: #{ "Accept": "application/json" },
});

// Follow + retry on 5xx
for attempt in 0..3 {
    let r = http("https://api.example.com/", #{ follow_redirects: true });
    if r.status < 500 { return r.status == 200 ? 0 : 1; }
    sleep_ms(1000 * (attempt + 1));
}
return 2;

// Proxy + HSTS
http("http://example.com/", #{
    proxy: "socks5h://127.0.0.1:9050",
    hsts: "/tmp/hsts.txt",

```

```
});

// mTLS
http("https://mtls.example.com/", #{
  client_cert: "/etc/keys/bundle.pem",
});

// Unix socket
let r = http("http://localhost/v1.40/version", #{
  unix_socket: "/var/run/docker.sock",
});
print(json_stringify(json_parse(r.body), 2));
```

Browser (sticky-session) binding

A stateful HTTP "browser" — cookies and headers persist across calls against the same handle. Matches the common "log in once, then make authenticated requests" pattern.

Function	Description
<code>browser()</code>	New browser with a tempfile cookie jar.
<code>browser(opts)</code>	With initial config.
<code>use_persistent_session(name)</code>	Browser method; jar is <code>~/.recon/jars/<name>.db</code> .
<code>
.get(url [, opts])</code>	GET via the browser.
<code>
.post(url, opts)</code>	POST.
<code>
.put(url, opts) / .patch(...) / .delete(url [, opts])</code>	Other methods.
<code>
.request(method, url, opts)</code>	Catch-all.
<code>
.header(name, value)</code>	Sticky request header.
<code>
.headers(map)</code>	Merge sticky headers.
<code>
.clear_headers()</code>	Remove sticky headers.
<code>
.user_agent(str)</code>	Override User-Agent for the lifetime of the browser.
<code>
.basic_auth(user, pass)</code>	Sticky Basic auth.
<code>
.timeout_ms(n) / .connect_timeout(secs)</code>	Sticky timeouts.

Function	Description
<code>
.insecure(bool) / .follow_redirects(bool) / .max_redirects(n)</code>	Sticky knobs.
<code>
.cookiejar()</code>	Current jar path.
<code>
.cookies()</code>	Array of cookie records from the jar.
<code>
.cookie_set(name, value, domain, path)</code>	Inject a cookie.

Examples

```
// Login, then fetch protected resources
let br = browser();
br.user_agent("recon-test/0.58");
br.post("https://example.com/login", #{
  form: #{ user: "alice", pass: "s3cr3t" },
});
let r = br.get("https://example.com/dashboard");
assert(r.status == 200, "dashboard fetch failed");

// Persist across runs – next invocation sees the same cookies
let br = browser();
br.use_persistent_session("gh-session");
br.header("Accept", "application/vnd.github+json");
let r = br.get("https://api.github.com/user");

// Multi-persona fan-out
let personas = [{ name: "a", jar: "persona-a" }, { name: "b", jar: "persona-b" }];
for p in personas {
  let br = browser();
  br.use_persistent_session(p.jar);
  let r = br.get("https://api.example.com/whoami");
  print(`${p.name}: ${r.status}`);
}
```

TCP, TCP-server, UDP

0.57.0 shipped the server bindings. Handles are Send+Sync so they survive `thread_spawn` (0.56.0) boundaries — the expected pattern is "accept on main, spawn a handler per connection".

TCP client (existing)

Function	Description
<code>tcp(url)</code>	TCP connect probe; returns <code>{ status, duration_ms, ... }</code> .
<code>tcp(url, opts)</code>	With <code>timeout</code> (ms).

TCP server (0.57.0)

Function	Description
<code>tcp_listen(addr)</code>	Bind. <code>addr</code> like <code>"0.0.0.0:8080"</code> or <code>"[::]:8080"</code> .
<code>tcp_accept(listener)</code>	Blocking accept. Returns a <code>TcpConn</code> .
<code>tcp_accept(listener, timeout_ms)</code>	Times out with an error.
<code>tcp_read(conn, n, timeout_ms)</code>	Read up to N bytes → <code>Blob</code> .
<code>tcp_read_line(conn, timeout_ms)</code>	<code>\n</code> -terminated line; CR/LF stripped.
<code>tcp_write(conn, blob str)</code>	Write all. Returns bytes.
<code>tcp_peer_addr(conn)</code>	Remote <code>SocketAddr</code> string.
<code>tcp_close(conn)</code>	Close connection.
<code>tcp_close_listener(l)</code>	Close listener.

UDP

Function	Description
<code>udp_bind(addr)</code>	Bind.
<code>udp_recv_from(sock, max_len)</code>	Block until a datagram arrives. Returns <code>{ data: Blob, addr: string }</code> .
<code>udp_recv_from(sock, max_len, timeout_ms)</code>	With timeout.
<code>udp_send_to(sock, blob str, addr)</code>	Returns bytes sent.
<code>udp_close(sock)</code>	Release.

Examples

```
// TCP client probe
let r = tcp("example.com:443");
print(`${r.status} in ${r.duration_ms}ms`);

// Concurrent TCP echo server (from script/tcp-echo.rhai)
let l = tcp_listen("127.0.0.1:9000");
loop {
    let conn = tcp_accept(l);
    thread_spawn(|c| {
        let peer = tcp_peer_addr(c);
        let line = tcp_read_line(c, 5000);
        tcp_write(c, `echo: ${line}` + "\n");
        tcp_close(c);
    }, conn);
}

// UDP listener (from script/udp-listen.rhai)
let s = udp_bind("127.0.0.1:9001");
loop {
    let r = udp_recv_from(s, 65536, 30000);
    print(`${r.addr} → ${r.data.len()} bytes`);
}

// UDP beacon sender
let s = udp_bind("0.0.0.0:0");
for i in 0..5 {
    udp_send_to(s, `beacon-${i}`, "127.0.0.1:9001");
    sleep_ms(200);
}
udp_close(s);
```

Threading bindings

0.56.0. Requires rhai's `sync` feature (enabled). `spawn` alone is reserved by Rhai — use `thread_spawn`.

Function	Description
<code>thread_spawn(fn_ptr)</code>	Spawn a closure. Returns a <code>ThreadHandle</code> .
<code>thread_spawn(fn_ptr, arg)</code>	With one forwarded arg.
<code>thread_spawn(fn_ptr, args_array)</code>	With N args.
<code>join(h)</code>	Block; returns the closure's return value (or the worker's error).

Function	Description
<code>tid()</code>	Current thread id (stable within a run).
<code>sleep(ms)</code>	Alias of <code>sleep_ms</code> .
<code>channel()</code>	Unbounded MPSC — returns <code>[sender, receiver]</code> .
<code>channel_bounded(n)</code>	Bounded; <code>try_send</code> returns false when full.
<code>send(tx, val)</code>	Blocking send.
<code>try_send(tx, val)</code>	Non-blocking; false on full.
<code>recv(rx)</code>	Blocking.
<code>recv(rx, timeout_ms)</code>	With timeout.
<code>try_recv(rx)</code>	Non-blocking; <code>()</code> when empty.

Examples

```

// Fan out probes, gather via channel
let c = channel();
let tx = c[0];
let rx = c[1];

let urls = [
    "https://a.example.com/",
    "https://b.example.com/",
    "https://c.example.com/",
];

for url in urls {
    thread_spawn(|u| {
        let r = http(u, #{ timeout_ms: 5000 });
        send(tx, `${u} → ${r.status}`);
    }, url);
}

for i in 0..urls.len() {
    print(recv(rx, 10000));
}

// Bounded channel – producer/consumer with back-pressure
let c = channel_bounded(4);
let tx = c[0];
let rx = c[1];

thread_spawn(|| {
    for i in 0..100 { send(tx, i); }
});

let sum = 0;
for j in 0..100 {
    sum += recv(rx, 1000);
}
print(`total: ${sum}`);

// Join for return value
let h = thread_spawn(|| {
    // Do something heavy
    let r = http("https://slow.example.com/");
    r.status
});
let status = join(h);
print(`completed with status ${status}`);

```

DNS binding

Function	Description
<code>dns(host)</code>	Default bundle: A, AAAA, MX, TXT, NS, CNAME, SOA. Returns a Map keyed by record type.
<code>dns(host, types_str)</code>	Custom types: "A,AAAA,MX" .
<code>dns(host, types_str, opts)</code>	<code>opts</code> may contain <code>servers: "1.1.1.1,8.8.8.8"</code> .

Examples

```
let r = dns("example.com");
print(`A: ${r.A.join(", ")}`);
print(`MX: ${r.MX.join(", ")}`);

// Custom resolver
let r = dns("example.com", "A,AAAA", #{ servers: "127.0.0.1:5353" });
print(r);

// Check DNSSEC chain presence
let r = dns("example.com", "DNSKEY,DS");
if r.DNSKEY.len() == 0 { print("no DNSKEY"); }
```

TLS probe binding

Function	Description
<code>tls(host [, opts])</code>	Handshake + cert introspection. Returns a Map with <code>subject</code> , <code>issuer</code> , <code>sans</code> , <code>not_before</code> , <code>not_after</code> , <code>days_remaining</code> , <code>fingerprints</code> , <code>chain</code> , <code>tls_version</code> , <code>cipher_suite</code> .

Examples

```

let t = tls("example.com:443");
print(`CN: ${t.subject}`);
print(`issuer: ${t.issuer}`);
print(`expires in ${t.days_remaining} days`);
if t.days_remaining < 14 { eprint("cert expiring soon"); return 1; }

// Verify an expected subject
let t = tls("api.example.com:443");
assert(t.sans.contains("api.example.com"), "SAN mismatch");

// Per-host SNI override
let t = tls("alt.example.com:443", #{ sni: "primary.example.com" });

```

Ping binding

Function	Description
<code>ping(host)</code>	Default TCP ping, 4 packets. Returns <code>#{ sent, received, loss_pct, min_ms, avg_ms, max_ms, ...}</code> .
<code>ping(host, opts)</code>	<code>opts</code> supports <code>count</code> , <code>icmp: true</code> , <code>timeout_ms</code> .

Examples

```

let r = ping("8.8.8.8");
print(`${r.received}/${r.sent} received, avg ${r.avg_ms}ms`);

// ICMP (needs privileges on macOS/Linux for non-DGRAM types)
let r = ping("example.com", #{ icmp: true, count: 10 });

// TCP ping to a specific port
let r = ping("example.com:443");
assert(r.received > 0, "no reachability");

```

File transfer bindings

FTP

Function	Description
<code>ftp(url)</code>	FTP / FTPS anonymous listing or retrieve.
<code>ftp(url, opts)</code>	<code>opts</code> : <code>user</code> , <code>pass</code> , <code>ftps_implicit</code> .

SFTP

Function	Description
<code>sftp(url)</code>	SSH-backed listing or retrieve.
<code>sftp(url, opts)</code>	<code>opts</code> : <code>user</code> , <code>pass</code> , <code>privkey</code> , <code>port</code> .

TFTP

Function	Description
<code>tftp(url)</code>	RFC 1350 download.
<code>tftp(url, opts)</code>	<code>opts</code> : <code>blksize</code> (RFC 2348).

Gopher

Function	Description
<code>gopher(url)</code>	RFC 1436 selector fetch.

Examples

```
let listing = ftp("ftp://ftp.example.com/");
print(`${listing.entries.len()} entries`);

let data = ftp("ftp://ftp.example.com/file.bin");
print(`${data.size} bytes`);

let h = sftp("sftp://me@example.com/srv/app.log", #{ privkey: "~/.ssh/ci_rsa"
file_write_all("/tmp/app.log", h.body);

let img = tftp("tftp://tftp.example.com/boot.img", #{ blksize: 8192 });
print(img.size);

let g = gopher("gopher://gopher.floodgap.com/0/gopher/proxy");
print_raw(g.body);
```

Mail retrieval bindings

POP3

Function	Description
<code>pop3(url)</code>	Capability + message listing.

Function	Description
<code>pop3(url, opts)</code>	<code>opts: stls</code> , <code>retrieve_index</code> (int, 1-based).

IMAP

Function	Description
<code>imap(url)</code>	EXAMINE + capabilities.
<code>imap(url, opts)</code>	<code>opts: fetch</code> (int — message index), <code>peek: true</code> .

Examples

```
// POP3 probe with STLS
let r = pop3("pop3://alice:s3cr3t@pop.example.com/", #{ stls: true });
print(`${r.message_count} messages`);

let first = pop3("pop3s://alice:s3cr3t@pop.example.com/", #{ retrieve_index:
print_raw(first.body);

// IMAP peek
let r = imap("imaps://alice:s3cr3t@imap.example.com/INBOX", #{ fetch: 3, peek
print_raw(r.body);
```

SMTP binding

Function	Description
<code>smtp(url)</code>	Capability + STARTTLS probe.
<code>smtp(url, opts)</code>	Send a message. <code>opts</code> keys: <code>from</code> , <code>to</code> (array), <code>subject</code> , <code>body</code> , <code>headers</code> , <code>auth</code> ("user:pass"), <code>helo</code> , <code>no_starttls</code> , <code>dkim_key</code> , <code>dkim_selector</code> , <code>dkim_domain</code> .

Examples

```
// Probe only
let r = smtp("smtp://mail.example.com:25");
print(`capabilities: ${r.capabilities.join(", ")}`);

// Send
smtp("smtp://smtp.example.com:587", #{
  from: "me@example.com",
  to: ["you@example.com"],
  subject: "Automated notice",
  body: "Probe result: OK",
  auth: env("SMTP_CREDS"),
});

// DKIM-signed
smtp("smtp://smtp.example.com:587", #{
  from: "me@example.com",
  to: ["you@example.com"],
  subject: "Signed",
  body: "This message is DKIM-signed.",
  dkim_key: "~/dkim/default.pem",
  dkim_selector: "default",
});
```

WebSocket binding

Function	Description
<code>ws(url)</code>	Handshake + Ping/Pong. Returns <code>#{ status, duration_ms, negotiated_protocol }</code> .
<code>ws(url, opts)</code>	<code>opts: headers, subprotocols, send_text, send_binary.</code>

Examples

```
let r = ws("wss://echo.websocket.events");
print(`connected in ${r.duration_ms}ms`);

// Send a frame, check response
let r = ws("wss://echo.websocket.events", #{ send_text: "hello" });
print(r.received);
```

Other protocol bindings

Brief interface reference; each has at least one example in `script/*.rhai`.

LDAP

```
let r = ldap("ldap://ldap.example.com");           // anonymous RootDSE query
```

RTSP

```
let r = rtsp("rtsp://camera.example.com/stream1");
```

Dict

```
let r = dict("dict://dict.org/d:recon");
```

NTP

```
let r = ntp("ntp://pool.ntp.org");  
print(`offset: ${r.offset_ms}ms, rtt: ${r.rtt_ms}ms`);
```

Memcached

```
let r = memcached("memcache://cache.example.com/?version");  
let s = memcached("memcache://cache.example.com/", #{ command: "stats" });
```

Redis

```
let r = redis("redis://cache.example.com/");       // PING  
let info = redis("redis://cache.example.com/", #{ command: "INFO server" });
```

MQTT

```
let r = mqtt("mqtt://broker.example.com", #{  
    topic: "sensors/room-1",  
    message: '{"temp": 21.3}',  
});
```

Encoding & decoding bindings

0.55.0 expanded encode with Aztec + PDF417 and added decode.

Function	Description
<code>encode::qr(text)</code>	PNG Blob.
<code>encode::datamatrix(text)</code>	PNG Blob.
<code>encode::barcode(format, text)</code>	PNG Blob for any 1D/2D.
<code>encode::encode(format, text)</code>	PNG Blob (default renderer).
<code>encode::encode(format, text, output_format)</code>	<code>output_format = ascii / svg / png</code> . Returns string for ascii/svg, Blob for png.
<code>encode::encode(format, text, output_format, opts)</code>	<code>opts: qr_level: "L" "M" "Q" "H"</code> .
<code>encode::decode(blob)</code>	Scan an image Blob → <code>{ text, format }</code> .
<code>encode::list()</code>	Supported format names.

Examples

```
// Generate QR
let png = encode::qr("https://example.com");
file_write_all("/tmp/site.png", png);

// Tune QR durability
let png = encode::encode("qr", "sticker-text", "png", #{ qr_level: "H" });

// Get SVG
let svg = encode::encode("qr", "recon", "svg");
file_write_all("/tmp/site.svg", svg);

// Decode
let data = file_read("/tmp/site.png");
let r = encode::decode(data);
print(`${r.format}: ${r.text}`);

// Round-trip test
let png = encode::qr("hello");
let r = encode::decode(png);
assert(r.text == "hello", "round-trip failed");
```

Hashing binding

Ten algorithms plus CRC32.

Function	Description
<code>md5(data)</code>	MD5 (hex string).
<code>sha1(data)</code>	SHA-1.
<code>sha224</code> / <code>sha256</code> / <code>sha384</code> / <code>sha512</code>	SHA-2 family.
<code>sha3_256</code> / <code>sha3_512</code>	SHA-3.
<code>blake3(data)</code>	BLAKE3.
<code>crc32(data)</code>	CRC32 (hex, 8 chars).
<code>hmac_sha256(key, data)</code>	Keyed HMAC.

`data` accepts string or Blob.

Examples

```
let h = sha256(file_read("big.iso"));
print(h);

// HMAC
let sig = hmac_sha256("sharedsecret", "payload");
print(sig);

// Hash an HTTP body
let r = http("https://example.com/");
let fp = blake3(r.body);
print(`fingerprint: ${fp}`);
```

Compression & archive bindings

Compression

Function	Description
<code>compression::compress(algo, data [, level])</code>	<code>algo</code> = <code>gzip</code> , <code>deflate</code> , <code>brotli</code> , <code>zstd</code> , <code>bzip2</code> , <code>lz4</code> , <code>xz</code> , <code>snap</code> .
<code>compression::decompress(algo, data)</code>	Reverse.
<code>compression::decompress_auto(data)</code>	Auto-detect from magic bytes.
<code>compression::list()</code>	Supported algos.

Archive

Function	Description
<code>archive::create(dest_path, sources_array)</code>	Infer format from extension.
<code>archive::extract(src_path, dest_dir)</code>	Extract zip/tar/tar.gz/tar.xz/tar.bz2.

Examples

```
// Round-trip
let body = http("https://example.com/page.html").body;
let gz = compression::compress("gzip", body, 6);
print(`${body.len()} → ${gz.len()} bytes`);
let back = compression::decompress("gzip", gz);
assert(back.len() == body.len(), "round-trip");

// Zip up
archive::create("/tmp/backup.zip", ["file1.txt", "dir1/"]);
archive::extract("/tmp/backup.zip", "/tmp/restore/");
```

Encryption bindings

age + PGP shell-out.

Function	Description
<code>encrypt::keygen()</code>	New X25519 age keypair. Returns <code>#{ public, private }</code> .
<code>encrypt::encrypt(data, recipients)</code>	<code>recipients</code> = array of age-pub strings.
<code>encrypt::decrypt(data, identity)</code>	<code>identity</code> = secret key string.
<code>encrypt::encrypt_passphrase(data, passphrase)</code>	Script-based.
<code>encrypt::decrypt_passphrase(data, passphrase)</code>	—

Examples

```

let kp = encrypt::keygen();
print(`public: ${kp.public}`);
file_write_all("~/age/identity", kp.private);

let plaintext = "highly classified";
let ct = encrypt::encrypt(plaintext, [kp.public]);
let pt = encrypt::decrypt(ct, kp.private);
assert(pt.to_string() == plaintext, "round-trip");

let ct2 = encrypt::encrypt_passphrase("hello", "correct horse battery staple")

```

Check-digit binding

Function	Description
<code>checkdigit::verify(algo, input)</code>	true / false.
<code>checkdigit::create(algo, partial)</code>	Computed full value.
<code>checkdigit::list()</code>	All 60+ algorithm names.

Examples

```

assert(checkdigit::verify("isbn13", "9780131103627"), "valid ISBN");

let full = checkdigit::create("ean13", "400638133393");
print(full);           // 4006381333931

for algo in checkdigit::list() {
    print(algo);
}

```

Sample-data binding

Function	Description
<code>sample::get(name)</code>	String content of a named sample.
<code>sample::list()</code>	Array of sample names.

Examples

```
let u = sample::get("json-user");
print(u);

for s in sample::list() { print(s); }
```

JWT binding

Function	Description
<code>jwt::view(token)</code>	Decode header + claims (no signature check).
<code>jwt::sign(payload_map, opts)</code>	opts: alg, secret or key path.
<code>jwt::validate(token, opts)</code>	opts: alg, secret or key / pubkey.

Examples

```
let t = jwt::sign({ sub: "alice", exp: now() + 3600 }, {
  alg: "HS256",
  secret: env("JWT_SECRET"),
});

let ok = jwt::validate(t, { alg: "HS256", secret: env("JWT_SECRET") });
assert(ok, "validation failed");

let parts = jwt::view(t);
print(`alg: ${parts.header.alg}, sub: ${parts.payload.sub}`);
```

Email-protection binding

Function	Description
<code>email::spf(domain)</code>	SPF record + validation.
<code>email::dmarc(domain)</code>	DMARC policy.
<code>email::dkim(domain, selector)</code>	DKIM record for selector.
<code>email::mta_sts(domain)</code>	MTA-STS policy.
<code>email::bimi(domain [, selector])</code>	BIMI record.
<code>email::tls_rpt(domain)</code>	SMTP TLS-RPT record.

Examples


```

let s = email::spf("example.com");
if s.valid { print("SPF OK"); } else { print(`SPF: ${s.error}`); }

let d = email::dmarc("example.com");
print(`policy: ${d.policy}`);

let dk = email::dkim("example.com", "selector1");
print(dk.record);

```

Netstatus binding

Function	Description
<code>netstatus::run()</code>	Execute the probe bundle defined in <code>~/.recon/config.toml</code> [netstatus] .

```

let r = netstatus::run();
for p in r.probes {
    print(`${p.name}: ${p.status}`);
}

```

Text-encoding binding

Function	Description
<code>text::decode(blob, charset)</code>	Decode bytes → string.
<code>text::encode(string, charset)</code>	Encode string → bytes.
<code>text::detect(blob)</code>	Auto-detect (<code>chardetng</code>). Returns charset label.
<code>text::normalize_newlines(s, style)</code>	style = <code>unix</code> / <code>windows</code> / <code>mac</code> .
<code>text::list_charsets()</code>	Supported labels.

Examples

```

let bytes = file_read("greek.txt");
let charset = text::detect(bytes);
let utf8 = text::decode(bytes, charset);
file_write_all("greek-utf8.txt", text::encode(utf8, "utf-8"));

let crlf = text::normalize_newlines("line1\nline2\n", "windows");

```

Whois binding

Function	Description
<code>whois(domain)</code>	Two-hop whois (registry then registrar). Returns raw text.

```
let w = whois("example.com");
print_raw(w);
```

IPFS binding

Function	Description
<code>ipfs(url)</code>	Fetch <code>ipfs://</code> or <code>ipns://</code> via the configured gateway.

```
let data = ipfs("ipfs://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG");
print(data.body.len());
```

Agent-browser binding

Exposed as a static module. Requires `agent-browser` on `$PATH`.

Function	Description
<code>agentBrowser::available</code>	Bool.
<code>agentBrowser::version</code>	Version string.
<code>agentBrowser::open(url)</code>	Navigate to URL.
<code>agentBrowser::close()</code>	End the session.
<code>agentBrowser::click(selector)</code>	Click by CSS selector or <code>@ref</code> .
<code>agentBrowser::fill(selector, text)</code>	Fill a form field.
<code>agentBrowser::type(selector, text)</code>	Type into a field (character events).
<code>agentBrowser::keyboard_type(text)</code>	Keyboard-level typing.
<code>agentBrowser::press(key)</code>	Key press (e.g. "Enter", "Tab").
<code>agentBrowser::screenshot(path)</code>	Save PNG.
<code>agentBrowser::pdf(path)</code>	Save PDF.
<code>agentBrowser::snapshot()</code>	Accessibility-tree dump.
<code>agentBrowser::eval(js)</code>	Execute JS, return the value.

Function	Description
<code>agentBrowser::get(selector)</code>	Read innerText / value.
<code>agentBrowser::find(selector)</code>	Locate (returns ref).
<code>agentBrowser::cmd(args_array)</code>	Raw passthrough to the CLI.

Examples

```
if !agentBrowser::available { return 2; }

agentBrowser::open("https://example.com/login");
agentBrowser::fill("#user", "alice");
agentBrowser::fill("#pass", "s3cr3t");
agentBrowser::click("button[type=submit]");
agentBrowser::screenshot("/tmp/after-login.png");
agentBrowser::close();
```

SQLite binding

Function	Description
<code>sqlite(spec) / sqlite(spec, mode)</code>	Open. spec = path, <code>":memory:"</code> , or an alias. mode = <code>ro</code> <code>rw</code> (default) <code>rwc</code> .
<code>.query(sql) / .query(sql, params)</code>	Array of Maps.
<code>.query_one(sql, params)</code>	First row as a Map (or <code>()</code>).
<code>.query_value(sql, params)</code>	Single scalar value.
<code>.exec(sql, params)</code>	Row count affected.

Examples

```
let db = sqlite(":memory:");
db.exec("CREATE TABLE users (id INTEGER PRIMARY KEY, name TEXT);");
db.exec("INSERT INTO users (name) VALUES (?)", ["alice"]);
db.exec("INSERT INTO users (name) VALUES (?)", ["bob"]);

let rows = db.query("SELECT id, name FROM users ORDER BY id");
for r in rows {
  print(`${r.id}: ${r.name}`);
}

let count = db.query_value("SELECT COUNT(*) FROM users");
print(`total: ${count}`);
```

Document-conversion bindings

0.58.0. comrak for HTML; agent-browser for PDF.

Function	Description
<code>md_to_html(md)</code> / <code>md_to_html(md, opts)</code>	Markdown (string or Blob) → HTML string.
<code>md_to_pdf(md, dest)</code> / <code>md_to_pdf(md, dest, opts)</code>	Markdown → PDF at <code>dest</code> . Needs agent-browser.
<code>html_to_pdf(html, dest)</code>	HTML → PDF at <code>dest</code> . Needs agent-browser.

Opts map

Key	Default	Description
<code>toc</code>	false	Inject linkable TOC.
<code>toc_depth</code>	3	Include H1..H N.
<code>toc_title</code>	"Contents"	—
<code>title</code>	document default	<code><title></code> + PDF metadata.
<code>css</code>	—	Additional CSS (appended after default).
<code>no_default_css</code>	false	Skip bundled default CSS.
<code>gfm</code>	false	GitHub-flavored extensions.

Examples

```
let md = file_read("README.md").to_string();

// md → html
let html = md_to_html(md, #{ toc: true, toc_depth: 3, gfm: true, title: "README" });
file_write_all("/tmp/readme.html", html);

// md → pdf
md_to_pdf(md, "/tmp/readme.pdf", #{ toc: true, gfm: true, title: "README" });

// html → pdf
let html = http("https://example.com/report.html").body.to_string();
html_to_pdf(html, "/tmp/report.pdf");

// Fully custom styling
md_to_pdf(md, "/tmp/styled.pdf", #{
  no_default_css: true,
  css: file_read("print.css").to_string(),
  toc: true,
  toc_title: "Table of Contents",
});
```

Part IV — Appendices

Exit codes

Curl-compatible exit codes for common cases:

Code	Meaning
0	Success.
1	Generic protocol error.
2	Source-load error (<code>--compare</code> , etc.).
3	Agent-browser / Chrome unavailable (PDF features).
6	DNS resolution failed.
7	Connection refused.
22	HTTP ≥ 400 with <code>-f</code> (or <code>--fail-with-body</code>).
28	Operation timeout (<code>--max-time</code>).
35	TLS handshake error.
47	Redirect limit exceeded.
52	Empty reply from server.
55	Send error.
56	Receive error.
60	Peer cert cannot be authenticated.

Script-mode exit codes:

- `return N` from the top-level script becomes the process exit code (mod 256).
- Unhandled script errors → exit 1 (or the last `ProtocolExitCode` tag).

Environment variables

Variable	Read by	Description
<code>HTTP_PROXY</code> / <code>http_proxy</code>	<code>--proxy</code>	Proxy for http:// URLs.

Variable	Read by	Description
HTTPS_PROXY / https_proxy	--proxy	Proxy for https:// URLs.
ALL_PROXY / all_proxy	--proxy	Fallback proxy.
NO_PROXY / no_proxy	--noproxy	Bypass list.
SSL_CERT_FILE	--cacert	Extra trust-root bundle.
CURL_CA_BUNDLE	--cacert	Same as SSL_CERT_FILE.
RECON_NO_PAGER	--help / --examples	Disable paging.
RECON_IPFS_GATEWAY	--ipfs-gateway	Default IPFS gateway.
RECON_CONFIG	config file	Override path to config.toml .
AGENT_BROWSER_JSON	—	When 1 , agent-browser returns JSON (set internally by recon).
EDITOR	--editor	Editor binary for response inspection.
HOME	various	Used to locate ~/.recon/ .
PAGER	--help / --examples	Override default less -FRX .

Configuration file

~/.recon/config.toml — commented skeleton created by recon --init .

Structure

```

# Default flag overrides
[defaults]
connect_timeout = 30
max_time = 60
user_agent = "recon/0.58.1 (auto)"

# Netstatus probe bundle
[netstatus]
probes = [
    { name = "dns", target = "8.8.8.8" },
    { name = "http", url = "https://example.com/" },
    { name = "tls", target = "example.com:443" },
]

# Per-host SNI → cert mapping for --serve-tls
[[serve_sni]]
host = "a.example.com"
cert = "/etc/certs/a.pem"
key = "/etc/certs/a.key"

[[serve_sni]]
host = "b.example.com"
cert = "/etc/certs/b.pem"
key = "/etc/certs/b.key"

```

~/.recon/ layout

Created by `recon --init`:

```

~/.recon/
├── config.toml           # Commented skeleton; overrideable via RECON_CONF
├── script/              # Bare-word scripts (`recon --script NAME`)
│   └── *.rhai
├── jars/                # Persistent cookie jars
│   └── <session>.db
├── sni/                  # Per-host TLS cert + key pairs for --serve-tls
│   ├── a.example.com/cert.pem
│   └── a.example.com/key.pem
└── hsts.txt             # Optional HSTS cache (curl-compatible)

```

Glossary

- **Sticky session** — a `browser()` handle whose cookies + headers persist across calls. Pair with `use_persistent_session` for cross-invocation persistence.

- **Script defaults** — the frozen snapshot of CLI flags (`-H`, `-k`, `--connect-timeout`, ...) that every script binding inherits when the caller doesn't override via an opts map.
 - **Bare-word script** — one that can be invoked by name because it lives in `~/.recon/script/`, e.g. `recon --script http` → `~/.recon/script/http.rhai`.
 - **Protocol exit code** — recon's internal scheme for passing an exit code through a panic-style error path from protocol bindings (so a failed SMTP probe can exit with curl's `CURLE_URL_MALFORMAT`, for instance).
 - **agent-browser** — external CLI that wraps Chrome DevTools Protocol. Used for `--browser-screenshot`, `--md-to-pdf`, `--html-to-pdf`, and the `agentBrowser::*` script bindings.
-

End of manual. For the curated-example companion, run `recon --examples`. For topic-specific deep dives, run `recon --help <topic>`.